

Marks 100

**CS4048-Data Science
Assignment No 3**

Due Date: March 19, 2026

Q1. Implement ANN based solution for the Classification of Rice varieties (Arborio, Basmati, Ipsala, Jasmine and Karacadag). Submit complete report, working code and readme file. (15 marks)

<https://www.kaggle.com/datasets/muratkokludataset/rice-image-dataset>
<https://www.sciencedirect.com/science/article/abs/pii/S0168169921003021?via%3Dihub>

b. Test the same solution for the classification of date fruits. You need to prepare another local dataset called NU-Dates and test the accuracy of the model. (15 marks)

<https://onlinelibrary.wiley.com/doi/10.1155/2021/4793293>
<https://onlinelibrary.wiley.com/doi/epdf/10.1155/2021/4793293>
<https://www.muratkoklu.com/datasets/>

Q2. Implement ANN based solution for the Customer Churn Prediction/Analysis problem. (30 marks)

<https://www.kaggle.com/code/richardcsuwandi/ann-on-bank-customers-dataset>
<https://www.kaggle.com/datasets/filippoo/deep-learning-az-ann>
<https://www.kaggle.com/code/richardcsuwandi/ann-on-bank-customers-dataset>

Q3. Implement ANN based solution for the detection of motorbike brand, cc & wearing of helmet. Create or search your own dataset. Testing will be on real data in FAST. (40 marks)